Binomial coefficient powers: an entire-function reduction and small-degree cases

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7 November 2026

Abstract

The Le Roy equivalence and all degrees at most five are proved. Arbitrary noninteger exponents at least two in degrees six and above remain unresolved here.

Catalogue scope. AIM 682; partial progress. Canonical target at revision `1e56463`. The binomial-power question is stated in [1, Section 4.5]. The classical multiplier-sequence normalization is that of [2, Theorem 6].

1 Statement and argument

Status: partial only; this problem is not counted as resolved. The target asks whether

$$B_{n,\gamma}(x) = \sum_{k=0}^{n} \binom{n}{k}^{\gamma} x^k$$

has only real zeros for every $n \geq 1$ and every real $\gamma \geq 2$. The following exact reduction identifies one remaining entire-function question. It does not settle that question for noninteger γ.

Proposition 1.1. *Fix $\gamma > 1$, and define the Le Roy entire function*

$$E_\gamma(z) = \sum_{k=0}^{\infty} \frac{z^k}{(k!)^\gamma}.$$

The following are equivalent:

1. *$B_{n,\gamma}$ has only real zeros for every $n \geq 1$;*
2. *E_γ belongs to the Laguerre–Pólya class of type I, with nonnegative Taylor coefficients;*
3. *every zero of E_γ is a negative real number.*

Proof. First observe that E_γ is entire. For fixed k,

$$\frac{\binom{n}{k}^\gamma}{n^{\gamma k}} \longrightarrow \frac{1}{(k!)^\gamma}, \qquad 0 \leq \frac{\binom{n}{k}^\gamma}{n^{\gamma k}} \leq \frac{1}{(k!)^\gamma}.$$

Dominated convergence for the power series, uniformly on every compact disk, gives

$$B_{n,\gamma}(z/n^\gamma) \longrightarrow E_\gamma(z).$$

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

All coefficients of $B_{n,\gamma}$ are positive and its constant term is one. Therefore real zeros, when all zeros are real, are strictly negative. The displayed local uniform limit proves (1)⇒(2).

The implication (2)⇒(3) follows from Hurwitz's theorem and $E_\gamma(0) = 1$. Conversely, the coefficient formula for the order of an entire function, together with Stirling's formula, gives

$$\operatorname{ord}(E_\gamma) = \limsup_{k\to\infty} \frac{k\log k}{\gamma\log(k!)} = \frac{1}{\gamma} < 1.$$

If (3) holds, Hadamard factorization therefore expresses E_γ as a genus-zero product over negative real zeros, with constant factor one. Finite products converge locally uniformly and have nonnegative coefficients. This proves (3)⇒(2).

For (2)⇒(1), the Pólya–Schur theorem says that

$$\delta_k = (k!)^{1-\gamma}, \qquad k \ge 0,$$

is a multiplier sequence, since $E_\gamma(z) = \sum \delta_k z^k/k!$. Let T be the corresponding coefficient multiplier. Apply T to $(1+x)^n$, reverse its coefficients, and apply T again. Each step preserves real-rootedness: the reversal sends each nonzero real root to its reciprocal, and the constant and leading coefficients remain nonzero. Direct calculation gives

$$\begin{aligned} T(1+x)^n &= \sum_{k=0}^{n} \frac{n!}{(k!)^\gamma (n-k)!}\, x^k, \\ T\left[x^n (T(1+x)^n)(1/x)\right] &= \sum_{k=0}^{n} \frac{n!}{(k!)^\gamma ((n-k)!)^\gamma}\, x^k \\ &= (n!)^{1-\gamma} B_{n,\gamma}(x). \end{aligned}$$

Thus every $B_{n,\gamma}$ has only real zeros. □

Proposition 1.2. *For every real $\gamma \ge 2$, the polynomial $B_{n,\gamma}$ has only negative real zeros when $1 \le n \le 5$.*

Proof. The degree-one assertion is immediate. In degree two, $B_{2,\gamma}(x) = x^2 + 2^\gamma x + 1$ has nonnegative discriminant for $\gamma \ge 1$, and both roots are negative. In degree three,

$$B_{3,\gamma}(x) = (1+x)\left(x^2 + (3^\gamma - 1)x + 1\right),$$

which also has only negative real roots for $\gamma \ge 1$.

For degree four put $A = 4^\gamma$, $B = 6^\gamma$. Dividing by x^2 and setting $y = x + x^{-1}$ gives

$$x^{-2} B_{4,\gamma}(x) = y^2 + Ay + (B-2).$$

The two roots in y are real and at most -2 provided

$$A \ge 4, \qquad D := A^2 - 4B + 8 \ge 0, \qquad D \le (A-4)^2.$$

The last condition is equivalent to $B \ge 2A - 2$. For $\gamma \ge 2$,

$$\frac{A^2}{B} = (8/3)^\gamma > 4, \qquad \frac{B}{A} = (3/2)^\gamma > 2,$$

so all three conditions hold. For every real $y \le -2$, the two roots of $x^2 - yx + 1 = 0$ are negative real numbers. This proves degree four.

For degree five put $A = 5^\gamma$, $B = 10^\gamma$. Then

$$B_{5,\gamma}(x) = (1+x)\left[x^4 + (A-1)x^3 + (B-A+1)x^2 + (A-1)x + 1\right].$$

After division of the bracket by x^2, the same substitution gives

$$y^2 + (A-1)y + (B-A-1).$$

Its discriminant is $D = A^2 + 2A + 5 - 4B$. Its roots are real and at most -2 provided $A \geq 5$, $D \geq 0$, and $D \leq (A-5)^2$, the latter being equivalent to $B \geq 3A - 5$. For $\gamma \geq 2$,

$$\frac{A^2}{B} = (5/2)^\gamma > 4, \qquad \frac{B}{A} = 2^\gamma \geq 4,$$

which proves these conditions. The four roots of the bracket are negative real numbers, and its remaining factor has root -1. □

The reduction leaves the negative-real-zero assertion for E_γ at arbitrary noninteger $\gamma \geq 2$ unresolved. Checking finitely many degrees, or invoking integer-parameter hypergeometric formulas, does not establish that assertion.

References

[1] J. Jalowy, Z. Kabluchko and A. Marynych, *Zeros and Exponential Profiles of Polynomials II: Examples*, Constructive Approximation (2026), Section 4.5, Theorem 4.7 and Remark 4.8. https://doi.org/10.1007/s00365-026-09775-2.

[2] G. Csordas and T. Forgacs, *Multiplier sequences, classes of generalized Bessel functions and open problems*, J. Math. Anal. Appl. **433** (2016), 1369–1389. https://arxiv.org/abs/1505.00355v2. DOI: https://doi.org/10.1016/j.jmaa.2015.08.047.